Q. Define linkage?

Linkage is defined genetically as the failure of two genes to assort independently. Linkage occurs when two genes are close to each other on the same chromosome

Q. What is the difference between karyotype and idiotype

Karyotype	Idiotype
A **karyotype** is the number and appearance of chromosomes in the nucleus of a eukaryotic cell. The term is also used for the complete set of chromosomes in a species or in an individual organism	the karyotype of a species may be represented diagrammatically, showing all the morphological features of the chromosome; such a diagram is known as Idiotype

Q. What is polygenic inheritance?

Condition when two or more genes influence the expression of one trait. Polygenic inheritance is controlled by multiple genes

Q. What is epistasis?

Epistasis is a phenomenon that consists of the effect of one gene being dependent on the presence of one or more modifier genes Type of polygenic inheritance. Where alleles at one gene locus can hide or prevent expression of alleles at second gene locus.

Q. What is linked gene?

:Genes that are found on the same chromosomes are called linked gene.

Q. What is physical mapping?

: A physical map is primarily base on the location of landmark along a DNA molecule.

Q. write Name of different mapping technique?

1 : Cytogenetic map 2 : In situ hyberdization 3 : Genetic mapping 4 : Restriction enzyme 5 : Radiation hybrid mapping.

Q. What is position independent?

It is a forward genetics. It started at the phenotype, determining which protein was involvednand getting to the gene through the protein.

Q. How gene – species oligonucleotide are prepared?

Oligonucliotide isolating the protein product of the gene. These genes are usually produced large amount of protein. Then gene – species oligonucleotide are prepared.

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Q. How polygenic inheritance are controlled by multiple genes?

When two or more genes influence the expression of one trait. Then polygenic inheritance is controlled by multiple genes.

Q. What is multifactorial?

Many of traits are effected by the person,s environment, so they are called multifactorial.

Q. Give example of polygenic trait?

1 : Stature 2 : Shape of face 3 : Finger prints

Q. How many genes are controlled the chicken combs shapes? And where these found?

Comb shapes are controlled by two genes .These genes are found on two different chromosomes. Gene 1 : R , Gene 2 : p.

Q. Short note on hair colour?

Hair color is controlled by alleles on chromosomes 3, 6, 10 and 18. When more dominant alleles appear in genotype, the darker will be the hair.

Q. What is role of positive epistasis?

Positive epistasis is occurring between beneficial mutations. It generates improvement in function.

Q. What is negative effect?

When two mutation has a smaller effect than expected from their effects when alone.It is called negative effect.

Q. How sign epistasis occur?

It occur when one mutation has the opposite effect when in the presence of another mutation.

Q. What is dominant epistasis?

It is fruit color in summer squash. Epistasis gene exerts is effect with the presence of a dominant allele.

Q. Write duplicate dominant epistasis and its shape?

fruit shape in shephered,s purse . A – or B - =heart shape . aa and bb = narrow shape

Q. What is epigenetics?

Its literally means above the genetics. Epigenetics trait is a stable and heritable phenotype resulting from changes in chromosomes without alternation in DNA sequences.

Q. Write molecular mechanism of epigentics?

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1 : Methylation of cytosine posttranslational modification to histone protein , include acetylation, phosphorylation .

Q. how many epigenetic modifications are there write their name?

Three types of epigenetic modifications are there these are

1) Cytosine methylation 2) Histones modification 3)Non-coding RNAs (siRNAs)

Q. describes histones in Histone modification?

Histones are the proteins that organize the genetic material. Have a high percentage of basic amino acids, which gives histones an overall positive charge, while DNA is negative charged.

Q. Which two enzymes involved in Histone accyitylation?

1) HAT: Histone acetyltransferase 2) HDAC: Histone deacetylase.

Q. what is role of histones in accetylation?

When histones are acetylated, chromatin is open and genes are potentially active. When histones not tagged, deacetylated, the chromatin condenses and genes silenced.

Q. what are Common Epigenetics regulated phenomena?

1) Cellular Differentiation 2) X Chromosome inactivation 3) Genomic imprinting

Q. Describe Identical twins?

Identical twins are from the same zygote, so they begin life with the same genetic information. While infants they experience the same or very similar environments, so there is little variation in the epigenome. Over time the twins' environments will diverge, resulting in individual epigenetic tags to form for each twin. The difference in the twins' epigenomes is what makes them become different when they are older. The epigenetic tags can have such an effect on the twins that one can develop a disease while the other is fine.

Q. what is Vorinostat ?

Vorinostat is a histone deacetylase inhibitor.

Q. what environmental factors affected by epigenetics?

Nutrition of mother, stress mother, social interactions and physical activities, Exposure to toxins and diet.

Q. defines penetrance?

Proportion of individuals carrying a particular variant of a gene (allele or genotype) that also expresses an associated trait (phenotype)

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Q. what is variable expressivity?

When phenotype is expressed to a different degree among individuals with the same genotype it is called variable expressivity.

Q. MEN 1 stand for?

Multiple endocrine neoplasia 1 it is a hereditary disorder

Q. Write any two types of penetrance?

Complete penetrance, highly penetrance, incomplete or reduced penetrance, low penetrance.

Q. What is chromosomal aberration?

Any change in normal number or structure of chromosomes. Affects individuals may be due altering the amounts of products of the genes involved. Altered amounts may cause anomalies directly or may alter the balance of genes

Q. How many types of chromosomal aberration are there describe them?

1) Numerical (usually due to de novo error in meiosis).

2) tructural (may be due to de novo error in meiosis or inherited).

Q. Which structural abnormalities may occurs due to change in chromosomes?

Structural abnormalities may occurs due to

1) Translocations 2) Deletions 3) Duplications 4) Inversions

Q. chromosomal identifications dependent upon which factors? Write their name?

1. Size 2. Location of the centromere 3. Banding patterns

Q. how many chromosome are present in down syndrome?

Ans: 47 chromosome

Q. Write name of Common numerical Autosomal and sex chromosome Abnormalities?

Ans Autosomal 1) Downsyndrome (Trisomy 21) 2) Edwards syndrome (trisomy 18) 3) patau syndrome (trisomy13)

sex chromosomes abnormalities 1) Turner syndrome 45,x 2) klinefelter syndrome 47,xxy

Q. wite the characteristics of Down syndrome

Ans it is a trisomy of chromosomes no 21. The frequency of trisomy increases with increasing maternal age. Sometimes, robertssonian translocation involving chromosome 21. 3-4 %, not related to maternal age.

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Q write characteristic of turner syndrome.

Ans Relatively common disorder caused y the loss of genetic material from one of the sex chromosomes. Affects only females and they have only one x chromosome.

Q. Define Triploidy ? how many chromosomes that deseas contain?

Ans This is phenomena in which all the set of chromosomes contain one extra chromosome like a normal person have 2 pair of chromosomes and triploidy person have 3 chromosomes so they become 69 chromosomes and a normal person has 46 chromosomes.

Q. Define structural abnormalities?

Ans rearrangement because of chromosome breakage and subsequent reunion in a different configuration.

what are the types of structural abnormalities.

Ans there are 6 types i.e 1) Trasnlocation 2) Deletions 3) insertions 4) inversions 5) ring chromosomes 6) isochromosomes

Q. Define Translocation? Write its name?

Ans Transfer of genetic material from one chromosome to another chromosome called translocation. There are two types of translocations 1) Reciprocal translocation and 2) Robertosnian Translocation.

Q. Defien Reciprocal Translocation?

Ans An exchange of genetic material between two different chromosomes is called a reciprocal translocation in which no loss or gain of chromosomal material.

Q. Define Robertsonian translocations.?

Ans Break occurs at the extreme end of the short arms of two non homologuous acrocentric chromosomes and the small acentric fragments are lost so the the larger fragments fuse at their centromeric regions to for a single chromosome. this type of translocation occurs between chromosomes no 13 and 14 in humans.

Q write common human diesease caused by translocations?

Ans the common disease in humans caused by translocations are 1) cancer 2) infertility 3) down syndrome 4) leukemia 5) XX male syndrome

Q. What is difference between disjunction and non-disjunction?

Ans separation of two homologous chromosomes during cell division is called disjunction. When two homologous chromosomes or sister chromatids fails to separate during cell division it is called as non-disjunction.

Q differentiate between meiotic and mitotic non disjuntion ?

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Ans if non-disjunction occurs during 1st meotic division then all the sperms/ova derived from primary spermatocyte /oocytes will be abnormal.

Mitotic non-disjuntion results in cells with differetn number of chromosomes due to breaking of the spindle fibers during metaphse or anaphase

Q. Explain the failure of chromosomes separation?

Ans 1) Failure of a pair of homologous chromosomes to separate in meisis1.

2) Failure of sister chromatids to separate during meiosis II.

3) Failure of sister chromatids to separate during mitosis.

Q what will be the results in Mitotic non-disjuntion?

Ans Mitotic non-disjunction results in trisomy and monosomy. it causes trisomic and monosomic daughter cells to be formed.

Q. What are the causes of Non-Disjunciton ?

Ans there are several cause of non-disjuntion but the most are 1) Aging effect on primary oocyte 2) radiation from the environment 3) delayed fertilization after ovulation

Q. What is the difference between Trisomy and Monosomy?

Trisomy is a type of polysomy in which there are three copies of a particular chromosome, instead of the normal two. Trisomy is a type of aneuploidy (an abnormal number of chromosomes).

Monosomy occurs when there is loss of a single chromosome. monosomy of autosomes is lethal during sex chromosome monosomy will be Turner syndrome XO in which loss of sex chromosomes and they don't survive and died

Q. Define Trisomy Types

. Ans There are 4 types of Trisomy. 1) Primary Trisomy - An entire etra chromosome has been copied it is called primary trisomy. 2) Partial trisomy- in which an extra copy of part or fragment of a chromosomes present called partial trisomy. 3) Secondary trisomy- the extra chromosome has duplicated arms so it is called secondary trisomy. 4) Tertiary trisomy - extra chromosome is made up of copies of arms from two other chromosomes

Q. What are the common Trisomy in humans?

Ans Trisomy 21 , Trisomy 18 trisomy 13 and Klinefelter's syndrome (47xxy)

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Q. What are the causes of Monosomy?

Ans during Non-disjunction one gamete receives two copies of homologous chromosomes and other will have no copy. loss of chromosome as it move towards pole of cell during anaphase.

Q. Write the name of Monosomy diseases in humans.

Ans Turner syndrome and Cri Du chat syndrome

Q Define polyploidy?

Ans Multiples of haploid cells or no of chromosomes are called polyploidy . it can be in form of triploidy, tetraploidy , pentaploidy etc. during polyploidy fetus does not survive this phenomena is seems in many crops.

Q. write types of polyploidy in plants.

Ans Triploid (3x) Watermelon, Tetraploid(4x) cotton , Pentaploid (5x) Kenai Birch , Hexaploid (6x) Wheat

Q. Write the causes of Polyploidy

Ans 1) Retention of polar body. 2) Formation of diploid sperms 3) Di-spermy- fertilization by two sperms.

Q. Write polyploidy examples in plants.

Ans Triploid - Apple , Bannana, citrus fruits , ginger watermelons .

Tetraploid. Cotton, patato, tobacco, peanut

Hexaploid - Oat, Kiwifruit

Octaploid- Straberry

Decaploid- Some sugar cane hybrids

Q. What is difference between Autopolyploidy and Allopolyploidy.

Ans Autopolyploids are polyploids with multiple chromosome sets dervied from a single species. in case of allopolyploids are polypoloids with chromosomes derived from different species in the result of cross between two different species. Interspecies crosses can generate alloploids and their offspring generally sterile.

Q. Which chemical is used to promote the polyploidy in plants?

Ans is Colchicine is used to promote polyploidy in the plants to avoid non disjunction.

Q. What does the sterile means?

Ans The word sterile means that the specie cannot reproduce.

Q. What is the difference between Aneuploidy and Euploidy?

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Ans Aneuploidy:- change in chromosomes number can occur by the addition /deletion of chromosome or part of a chromosome. In case of Euploidy :- gain of one or more complete sets of chromosomes

Q. what is the difference between Terminal and interstitial deletions?

Ans in the Terminal the part of the chromosomes that is at the end like tips of the chromosomes been deleted and in interstitial the part of chromosomes which is at middle part or two breaks.

Q. Define Gene or chromosomal duplication?

ANs Gene duplication or chromosomal duplication is a major mechanism through which new genetic material is generated during molecular evolution

Q. Define Replication Slippage?

Ans Replication slippage is an error in DNA replication that can produce duplication of short genetic sequences. during replication DNA polymerase begins to copy the DNA.

Q. Write the causes of chromosomal duplications?

Ans 1) DNA Replication - Gene duplications can arise as products of several types of errors in DNA replication and repair machinery. 2) Ectopic homologous recombination;- common sources of gene duplications include ectopic homologous recombination 3) ectopic recombination Misaligned homologous - Duplicatoin arise from an event termed unequal crossing-over that occurs during meiosis between misaligned homologous chromosomes. 4) others causes of duplications - Retro-transposition event, aneuploidy, polyploidy, and replication silpage are also cause of duplications

Q. Define Dispersed and Tandem Duplications.

Ans. Some duplications are dispersed found in different locations from each other. Tandem duplications found next to each other

Q What is called pseudogenes?

Ans pseudogenes are inactive forms of gene forms by mutations, these genes cannot produce messenger RNA.

Q. What is called Chromosomal inversions?

Ans it involves two break re-arrangement and segment is reversed there are two types Pericentric inversions in which centromere is involved and paracentric in which only chromosomes Arms like P or Q is involved.

Q. What are the chromosomal inversions affects?

Ans it affects gene expression gene is re-positioned in a way that alters its gene expression 2 % of the human population carries karyotipically detectable inversions.

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Q. Define Hetrozygote inversion?

Ans individual with one copy of a normal chromosome and one copy of an inverted chromosome called inversion heterozygotes usually phenotypically they are normal.

Q. what are the effects of chromosomal inversion in humans?

Ans The most common inversion seen in humans is on chromosome 9 , at inv 9 (p12q13) this inversion is generally considered to have no harmful effects but increased risk of miscarriage or infertility.

Q. Differentiate between ring chromosome and Philadelphia chromosome?

Ans **A ring chromosome** is a chromosome whose arms have fused together to form a ring. in humans it is denoted by small r and in drosophila it is denoted by capital R.

philadelphia chromosome:- is a specific chromosomal abnormality that is related with chronic myelogenous leukemia(CML). reciprocal translocation between chromosome 9 and chromosome 22 which is specifically designated t(9:22)(q34:q11)

Q. Mutations are source of genetic variation? Justify

Mutation and variation Mutations are a major source of genetic variation in a population increasing biodiversity. Some variations may help to survive better and longer. Mutations in gametes (egg & sperm) are passed onto offspring. Mutations in body cells only affect the organism in which they occur and are not passed onto offspring.

Q. write name of major types of mutations.

Major types of mutations

1) Points Mutations 2) Variations in chromosomes number 3) Variations in chromosomes structure

Q. writes differences between Somatic and Germline mutation

Somatic mutation: Arise in the somatic cells. Passed on to other cells through the process of mitosis. Effect of these mutations depends on the type of the cell in which they occur.

Germline mutation: They occur in the cells that produce gametes. This type of mutation usually passed on to future generation. In multicellular organisms, the term mutation is generally used for germ line mutations..

Q. Explain - Forward, Reverse and Neutral mutations

Wild type most common phenotype in natural populations of the organism is called as wild type phenotype. The effect of mutation is considered with reference to wild type phenotype.

Forward mutation: Mutation that alters the wild type phenotype into mutant phenotype.

Reverse mutation: Mutation that changes a mutant phenotype back into the wild type.

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Neutral mutation: Mutation that alters the amino acid sequence of the protein but does not change its function as replaced amino acid is chemically similar or the affected AA has little influence on protein function..

Q What is difference between loss of function of gain of function ?

Loss of function, Gain of function Loss of function: Complete or partial loss of the normal function. Structure of protein is so altered that it is no longer works correctly. Mutation can occur in regulatory regions that affect transcription, translation or splicing of the protein. Frequently recessive

Gain of function: Produces an entirely new trait. Such a mutation causes a trait to appear in inappropriate tissues or at inappropriate times in development. It is frequently dominant.

Q. Differentiate between conditional and lethal mutants?

Conditional mutation: Expressed only under certain conditions. E.g. A temperature-sensitive mutation can cause cell death at high temperature, but might have no deleterious effects at a lower temperature.

Lethal mutation: Causes the death of the organism. Death does not have to occur immediately, it may take several months or even years. Longevity of an individual is significantly reduced.

Q. write difference between spontaneous and induced mutations?

Spontaneous mutation: Mutations that result from natural changes in DNA.

Induced mutation results from changes caused by environmental, chemicals and radiations.

Q. Define mutagens?

Mutagen: Any environmental agent that increases the rate of mutation above the spontaneous is called a mutagen such as chemicals & radiations, X-rays, Mustard gas etc.

Q. Define Base analogs?

Base analogs: Chemicals with structures similar to that of any of the four standard bases of DNA. DNA polymerases cannot distinguish these analogs. They may be incorporated into newly synthesized DNA molecules.

Q. Define Point mutations:

Point mutations affect single sites on DNA and it is Substitution of one base for another. Or Deletion/addition of a single base or small number of bases.

Q. what is difference between Transition and transversion mutation.

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if purine (A/G) is replaced by purines and pyrimidine (T/C) is replaced by pyrimidines it is called transitions mutation.

If purine substitutes for pyrimidine or vice versa is called transversion.

Q. Define Silent Mutation

Silent Mutation: These types of mutations have affected the DNA sequence but not the protein. They have no effect on the organism's phenotype. There is change at DNA level but no change in protein level.

Q. Define Missense mutation and its effects on protein formation?

Missense mutation: Missense mutations substitute one amino acid for another. Some missense mutations have very **large effects**. Some have minimal or no effect. It depends on where the mutation occurs in the protein's structure and how big a change in the type of amino acid it is. Missense mutation produces a change in amino acid sequence in protein product (e.g. Histidine in for Arginine). May change function of protein or may not!

Q. Define nonsense mutation and its effects on protein formation

Nonsense mutation: Nonsense mutations convert an amino acid into a stop codon. **The effect** is to shorten the resulting protein. Sometimes this has only a little effect, as the ends of proteins are often relatively un-important to function. However, often nonsense mutations result in completely non-functional proteins.

Q. Define Addition mutation or frame shift mutations

Additions mutation: Addition mutation is actually results in a frame shift mutation. Often, it has severe effects on protein function. It is also called as insertions.

Q. Define Deletion Mutations and its effects.

Deletions mutations:- Deletion of one or more nucleotides in DNA sequence is called deletion mutations. These types of mutations have severe effects on the phenotype.

Q. How mutations alter gene function

Null allele (loss of function) - no gene product.

Hypomorph – decreased activity.

Gain of function - increased activity.

Dominant negative-antagonizes normal product

Neomorph - novel activity

Predicting that a gene product won't do the job

Deletion, nonsense, frameshift of sequence is deleterious.

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Mutation in splice site usually has sever effects.

Missense mutations depend on location in protein.

Q. Explain the Mutation nomenclature

All variants should be described at the most basic level, i.e. DNA level.

Descriptions should always be in relation to a reference sequence, either a genomic or a coding DNA sequence.

Although theoretically a genomic reference sequence seems best, in practice a coding DNA reference sequence is preferred.

Good for promoters, alternative splicing etc.

When the entire genomic sequence is not known, a cDNA reference sequence should be used.

To avoid confusion in the description of a variant it should be preceded by a letter indicating the type of reference sequence used.

Several different reference sequences can be used.

"c." for a coding DNA sequence (like c.76A>T) "g." for a genomic sequence (like g.476A>T)

"m." for a mitochondrial sequence (like m.8993T>C,) "r." for an RNA sequence (like r.76a>u)

"p." for a protein sequence (like p.Lys76Asn)

Describing genes / proteins, only official HGNC (HUGO gene nomenclature committee) gene symbols should be used (www.genenames.org)

DNA reference sequence used should preferably be from the RefSeq database, listing both database accession and version number (like NM_004006.2)

There is no nucleotide 0 (zero).

Nucleotide 1 is A of the ATG-translation initiation codon.

The nucleotide 5' of the ATG-translation initiation codon is -1, the previous -2, etc. e.g. in 5' UTR c.-12G>A

The nucleotide 3' of the translation stop codon is *1, the next *2, etc. e.g. in 3' UTR c.*70T>A

Nucleotide numbering is purely arbitrary and starts with 1 at the first nucleotide of the database reference file.

No +, - or other signs are used.

The sequence should include all nucleotides covering the sequence (gene) of interest and should start well 5' of the promoter of a gene.

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Q. Define Genetic Disorders and Its classification.

Genetic Disorders: A genetic disease or disorder appears as a result of mutations in an individual's DNA. Change in DNA sequence may be small or large like deletions/addition of entire chromosome.

Fully penetrant and low penetrant

Fully penetrant, where other genes and environmental factors have no effect

Low-penetrance, genes play a small part, along with other environmental factors

Classification of genetic disorders

1) Multifactorial disorders **2) Single gene disorders** **3) Chromosomal disorders**

4) Mitochondrial disorders **5) Somatic mutations (cancers)**

Q. Explain Inheritance patterns of genetic disorders

Autosomal Dominant: Affected individuals are heterozygote. One allele is mutated, while other one is normal.

Autosomal recessive: Homozygous, when both alleles of the gene are mutated.

X-linked recessive: Mutated alleles are present on either of the sex chromosomes.

Multifactorial: Genes and environment both contribute. Multifactorial frequent, single gene disorders less frequent: Multifactorial are common. Single gene less common like dominant/recessive pedigree patterns

Chromosomal disorders:- Thousands of genes may be involved. Multiple organ systems affected at multiple stages in gestation

Q. Explain Autosomal recessive inheritance / diseases

Autosomal Recessive diseases: Diseases occurs in individuals with two mutant alleles. In general, an individual inherit one mutant allele from each of the parent.

Recessive diseases: Homozygotes with two copies of the altered gene

Recessive diseases: Since each parent has two alleles. Chance of inheriting a mutant allele from one parent is 50% and for the other parent is also 50%. Net chance of inheritance of two mutant alleles is 25%.

Generally, the disease appears in the progeny of unaffected parents.

Affected progeny include both males and females equally.

Phenotypic proportions are equal – autosomal recessive: When we know that both male and female phenotypic proportions are equal, we can assume that we are dealing with autosomal inheritance.

Q. Write name of Common recessive disorders

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1) Bloom syndrome	2) Carpenter syndrome	3) Cystic fibrosis
4) Thalassemia	5) Many forms of mental retardation	6) Gaucher's disease
7) Glycogen storage diseases	8) Rotor syndrome	9) Many of eye diseases

Q. Properties of Autosomal recessive inheritance

Consanguinity: These diseases appear where parents have common ancestors.

Same allele increases chances of inheriting disorders: The chance of inheriting identical alleles by an individual from both parents increases the chance of inheriting a recessive disorder.

Common genetic background: Small populations of individuals with a common genetic background may have increased risk of recessive diseases. Ashkenazi: Gaucher disease. Phenotype found more likely in siblings of proband than in other relatives. Males and females are equally affected. In most of the cases, parents of affected individuals are asymptomatic. Risk for each of the sibling being affected. Risk of disease for each sib is 25 %.

Q. Explain Autosomal dominant inheritance/ disorders

Autosomal dominant disorders: In autosomal dominant disorders, the normal allele is recessive and the abnormal allele is dominant. In a typical pedigree, every affected person has one affected parent. Equal numbers of affected females/males are expected. Male to male transmission is possible. One-half of the children of an affected individual are expected to have inherited the dominant allele.

Autosomal Dominant disorders

Pedigree analysis, main clues for identifying a dominant disorder is that the phenotype tends to appear in every generation of the pedigree. Affected fathers/mothers transmit the phenotype to both sons and daughters. Trait is common in the pedigree. Trait is found in every generation.

Autosomal Dominant diseases

1) Marfan syndrome	2) Huntington's disease	3) Retinoblastoma
4) Waardenburg syndrome	5) Myotonic dystrophy	6) Polycystic kidney disease
7) Achondroplasia	8) Polydactyly	9) Heredity hearing loss

Q. Explain Dominance and Recessiveness? Define Hemizygous

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Dominance and recessiveness are the properties of characters/traits, not genes. A character is dominant if it is expressed in the heterozygote and recessive if not.

Recessiveness: Cystic fibrosis is recessive because only homozygotes manifest it, whereas heterozygotes show the normal phenotype.

Hemizygous: Males are hemizygous for loci on the X and Y chromosomes, where they have only a single.

Q. Explain X-linked recessive inheritance and disorders? Their Characteristics and disease.

X-linked recessive: Traits which are due to the genes which are present on either of the sex chromosomes. Many of the genes have a disease phenotype.

Males are hemizygous and express the disease: Males are hemizygous, will express the disease phenotype if one mutation is present.

Homozygous Females manifest disease: Females may be homozygous or heterozygous. Homozygous may manifest the disease.

General characteristic

1) Trait is rare in pedigree. 2) Trait skips generations. 3)Affected fathers do not pass to their sons.

4) Males are more often affected than females.

X-linked recessive disorders:-X-linked recessive: X-linked recessives expressed in all males but only in homozygous females.

Common X-linked disorders

1) Duchene Muscular Dystrophy 2) Hunter's Disease 3) Menkes Disease

4) Hemophilia A and B 5) Color Blindness

Q. Write name of Common X-linked dominant diseases

1) Rett syndrome 2) Alport syndrome 3) Goltz syndrome 4) Fragile-X Syndrome

Q. Explain maternal inheritance of mitochondrial DNA

Mitochondrial DNA (MTDNA) is inherited through ovum not sperm. Mother could pass it to all children. Father will pass it to none of the children. More than one copy of mtDNA is passed. Mutations are common.More than one type of mtDNA genome is passed.

Q. What is difference between mtDNA and Nuclear DNA ?

Nuclear DNA is inherited from all ancestors while mtDNA is inherited from a single lineage i.e from mother only.

Q. Write the Properties of Mitochondrial inheritance

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1) Heteroplasy 2) Variable expression 3) Pleiotropy

4) Reduced penetrance 5) Functional somatic mosaic

Q. writes the name of Common mitochondrial diseases.

1) Myoclonic epilepsy 2) Mitochondrial recessive ataxia 3) Leber hereditary optic neuropathy

4) Sensory ataxia neuropathy

Q. What is Multifactorial inheritance and disorders?

Multifactorial diseases: Genetic disorders controlled by many genes plus the effects of the environment.

Q. What is difference between genetic and environmental disorders in Multifactorial?

They are rare, genetically simple, unifactorial and high recurrence rate while environmental they are common genetically very complex, multifactorial and low recurrence rate.

Q. writes the name of Multifactorial diseases

1) Asthma 2) Cancers 3) Ciliopathies 4)Cleft palate 5) Diabetes 6) Heart diseases

7) hypertension 8) inflammatory bowel disease 9) intellectual disability

10) mood disorder 11) Obesity 12) infertility

Q. Explain Diagnostic tests for genetic disorders

Physical analysis: Physical analysis for affected individual.

Ultrasonography of fetus: Ultrasonography of fetus determines malformations of head, internal organs, and extremities.

Analysis of amniotic fluids: Amniotic fluid analysis to determine genetic and chromosomal disorders after 14 wks gestation. More than 200 various genetic diseases easily be detected.

Maternal blood analysis: Maternal blood analysis to observe abnormal fetus

Q. Define pedigree and its use.

A complete, detailed family history is called a pedigree. Pedigrees are used to determine the pattern of inheritance of a genetic disease within a family.

Q. Define Gene therapy

Gene therapy a procedure that involves identification, manipulation, and transference of genetic segments into a host to replace defective genes and to perform desired genetic activities.

Q. Explain Complications in basic inheritance pattern

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Single genes -Non-mendelian inheritance

Differences in gene actions can generate more complex inheritance patterns.

Single gene can give codominance, incomplete dominance, overdominance.

Complications to inheritance patterns

Multiple genes - epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance

Complications

1) A common recessive inheritance can give dominant pattern
2) Autosomal dominant inheritance with non-penetrance
3) Autosomal dominant inheritance with variable expression.

Q. What are the Goals/Reasons of pedigree analysis?

1) Determine the mode of inheritance: dominant, recessive, sex-linked.

2) Determine the probability of an affected offspring for a given cross

Q. Define DNA polymorphism and its phenotypic effects.

DNA polymorphism is a sequence difference compared to a reference standard present in at least 1–2% of a population. DNA Polymorphism may be single nucleotide or thousands of BASES. DNA Polymorphism may or may not have phenotypic effects

Q. Polymorphic sequences can serve as markers? Justify

If the location of a polymorphic sequence is known, it can serve as a marker for locating other genes or genetic regions. Each polymorphic marker has different versions or alleles. Polymorphisms can be silent, or be exhibited at levels of Proteins.

linked SNPs outside of gene :- have no effect on protein production or function.

Causative SNPs in gene:-

Non coding SNP: change amount of protein produced.

Co:ding SNP change amino acid sequence

Q. why to study- Polymorphism

Essential to study inheritance patterns. Map phenotypes and anchor genes to the genetic map by co-segregation analysis. Determine change in function: resistant/sensitive populations genetically determined variability among humans is due to a difference in 0.1% of the genomic sequence.

Q. what are the Types of Polymorphic DNA Sequences

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RFLP: Restriction Fragment Length Polymorphism

VNTR: Variable Number Tandem Repeats (8 to > 50 base pairs)

STR: Short Tandem Repeats (1– 8 base pairs).

SNP: Single Nucleotide Polymorphisms

AFLP: Amplified Fragment Length Polymorphism

Q. write the Applications of molecular markers

– short range

1) Parentage determination 2) Genetic distance estimation 3) Determination of twin zygosity 4) Sexing of pre-implanted embryos 5) Identification of disease carriers

– Long Range

1) Gene mapping & mapping of QTL by linkage 2) Markers assisted selection of breeds

Q. Explain Restriction Fragment Length Polymorphism and their uses in detailed

What is RFLP: RFLP is a difference in homologous DNA sequences that can be detected as fragments of different lengths after digestion of the DNA samples.

Most RFLP markers are heterozygous and highly locus-specific.

RFLP probe is a labeled DNA sequence that hybridizes with one or more fragments of the digested DNA sample after they were separated by gel electrophoresis.

Methodology

- Genomic DNA digested with restriction enzymes.
- DNA fragments separated via electrophoresis and transfer to nylon membrane.
- Membranes exposed to probes labeled with P32.
- Film exposed to X-Ray

Uses

- RFLPs can be used in paternity cases or criminal cases to determine the source of a DNA sample. (i.e. it has forensic applications).
- RFLPs can be used to measure recombination rates which can lead to a genetic map with the distance between RFLP loci.
- Direct method for selecting desirable genes such as disease resistance genes

RFLP used to determine disease status

- Determine disease status of an individual, i.e.

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- Huntington
- Cystic fibrosis
- Sickle cell anemia

Mutation detection

- RFLPs can be used to determine the disease status of an individual; (e.g. it can be used in the detection of particular mutations)

Q. Define VNTR

VNTR are short nucleotide sequences organized as a tandem repeats on genomes. Found on many chromosomes Show variations in length between individuals.

Each variant acts as an inherited allele, allowing them to be used for personal or parental identification.

Q. Write the Importance of VNTR

VNTRs are an important source of RFLP genetic markers used in linkage analysis (mapping) of genomes. When removed from surrounding DNA by the PCR or RFLP methods, and their size determined by gel electrophoresis or Southern blotting. As a result, they produce a pattern of bands unique to each individual.

Q. writes the Applications of VNTR

1) Microbiology 2) DNA fingerprinting 3) Genetic diversity

4) Forensics 5) Mapping of genomes 6) Breeding patterns of wild or domesticated animals

Q. Explain the Difference between AFLP & RAPD

Amplified Fragment Length Polymorphism	Random Amplified Polymorphic DNA
Restriction endonuclease digestion of DNA	PCR based marker with 10-12 base pairs
Ligation of adaptors	Random amplification of several fragments
Amplification of ligated fragments	Amplified fragments run in agarose gel detected by ethidium bromide.
Separation of the amplified fragments via electrophoresis and visualization	Unstable amplification leads to poor repeatability
AFLPs have stable amplification and good repeatability	

Q. Write uses of Molecular Markers

1) Different types of markers used to study diseases 2) DNA forensics

3) QTL traits in animals and plants

Q. Explain SNP- Screening and its importance

People can be grouped based on SNP profile

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1) Genome of each individual contains distinct SNP pattern.

2) People can be grouped based on the SNP profile.

SNP - found in coding and noncoding regions

1) SNPs are found in coding and (mostly) noncoding regions.

2) Occur with a very high frequency about 1 in 1000 bases to 1 in 100 to 300 bases.

SNP SCREENING methods: Two different screening strategies

1) Many SNPs in a few individuals 2) A few SNPs in many individuals

SNP Importance

1) In disease diagnosis 2) In finding predisposition to diseases 3) In drug discovery and development

4) In drug responses 5) Investigation of migration patterns 6) Medication and diagnosis at individual level

Q. What is the Frequency of autosomal dominant disorders

Frequency of autosomal dominant disorders varies between 3 – 9.5 per 1000 individuals

Q. What is the Location of intronic sequence with respect to exons

Beginning of intron

The number of the last nucleotide of the preceding exon, a plus sign and the position in the intron, like c.88+2T>G

End of intron

The number of the first nucleotide of the following exon, a minus sign and the position upstream in the intron, like c.89-1G>T

Q. What is the function of STRP

STR is simple repeats of nucleotide sequences.

1) Mononucleotide, dinucleotide 2) Used in linkage analysis 3) Used in DNA forensics

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